

ECG Noise and Lead-Failure Stress-Test Corpus (v2) — Generation Report and Full Record Roster

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Date: 2026-07-16 · **Repository:** <https://github.com/depolarised/artefaux> (tag v2.0.0) · **Sources** (every value below traces to these): `recipes/corpus.yaml`, `recipes/source_ids/*.csv`, `manifest.json` (all committed), regenerated in-memory from `artefaux.corpus + artefaux.recipes` (master_seed = 20260713). Generator: `scripts/reports/generate_c`

Scope. This is a *stress-test set*, not a *clinically representative cohort*. It deliberately over-samples failure; do not use it to estimate real-world prevalence or deployment accuracy.

1. Objectives

This is a deterministic generator and reproducible corpus **definition** for stress-testing ECG signal-quality gates (the internal `signalguard`) and noise detectors (`noiseguard`). It does **not** redistribute derived signals: it ships code, recipes, labels, manifest, and provenance so a user regenerates the corpus **bit-exactly** from their own open PhysioNet copies.

This report documents the generation algorithm and enumerates **all 85 stress records** with their corruption truth and expected behaviour. Records resolve against a local PhysioNet mirror; the concrete PTB-XL parent ids shown here are those in `recipes/source_ids/` (reproducible, seeded selection).

2. Corpus composition

Group	n	Source	Corruption
Naturally poor	15	PTB-XL technical-validation quality flags	none (inherently noisy)

Group	n	Source	Corruption
Real-noise pairs	30	clean PTB-XL + NSTDB <code>em/ma/bw</code>	SNR ladder $\{-6, -4, -2, +0, +2\}$ dB
Engineering extremes	40	clean PTB-XL parents (+ MACECGDB motion)	single-lead (13) + multi-lead (27)
Total	85		22 labelled data-integrity failures

Each record carries three label layers — **clinical parent** (authored rhythm class + PTB-XL quality flags; the Uni-G statement field is reserved and empty in v1), **corruption truth** (electrodes/leads, requested + measured SNR, noise segment, seed, α , amplitude bookkeeping), and **expected behaviour** (per-lead and record-level, mapped to the `signalguard` `QualityLabel` / `RecordQuality` and `noiseguard` discard vocabularies).

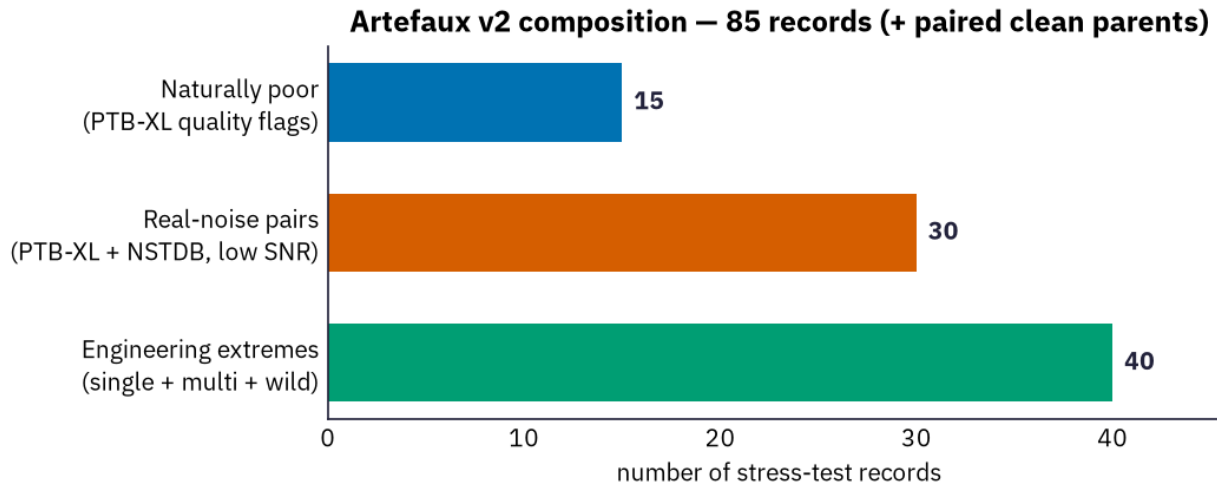


Figure 1: Corpus composition — 85 stress records plus paired clean parents.

3. Sources and licensing

Dataset	Role	Version	License
PTB-XL	clean + naturally-poor parents (500 Hz)	1.0.3	CC-BY-4.0
PTB-XL+	Uni-G/Glasgow layer — <i>cited and pinned, not consumed by v2</i>	1.0.1	CC-BY-4.0
MIT-BIH NSTDB	<code>em/ma/bw</code> noise for the SNR ladder	1.0.0	ODC-BY-1.0
MACECGDB	real stand-ing/walking/jumping motion	1.0.0	ODC-BY-1.0

PTB-XL, PTB-XL+, and MACECGDB are read from a local PhysioNet mirror; **only NSTDB** is fetched by `make download`. **Code** is GPL-3.0-or-later; **manifest, labels, and docs** are CC-BY-4.0. No source signals are redistributed — see `ATTRIBUTION.md`.

4. Generation algorithm

4.1 Canonical form. Every parent becomes a (12, T) float64 mV array in canonical lead order [I, II, III, aVR, aVL, aVF, V1-V6] at 500 Hz (NSTDB resampled from 360 Hz by deterministic polyphase resampling). Absent limb leads are derived from I and II.

4.2 Real-noise SNR mixing. For a clean lead x , real noise n , and requested SNR s (dB): $\alpha = \sqrt{P_x / (P_n \cdot 10^{s/10})}$, $y = x + \alpha n$, with P_x , P_n the DC-removed mean-square powers over the contaminated interval only. Both the requested and the *measured* SNR, and the exact NSTDB segment offset, are recorded in the label.

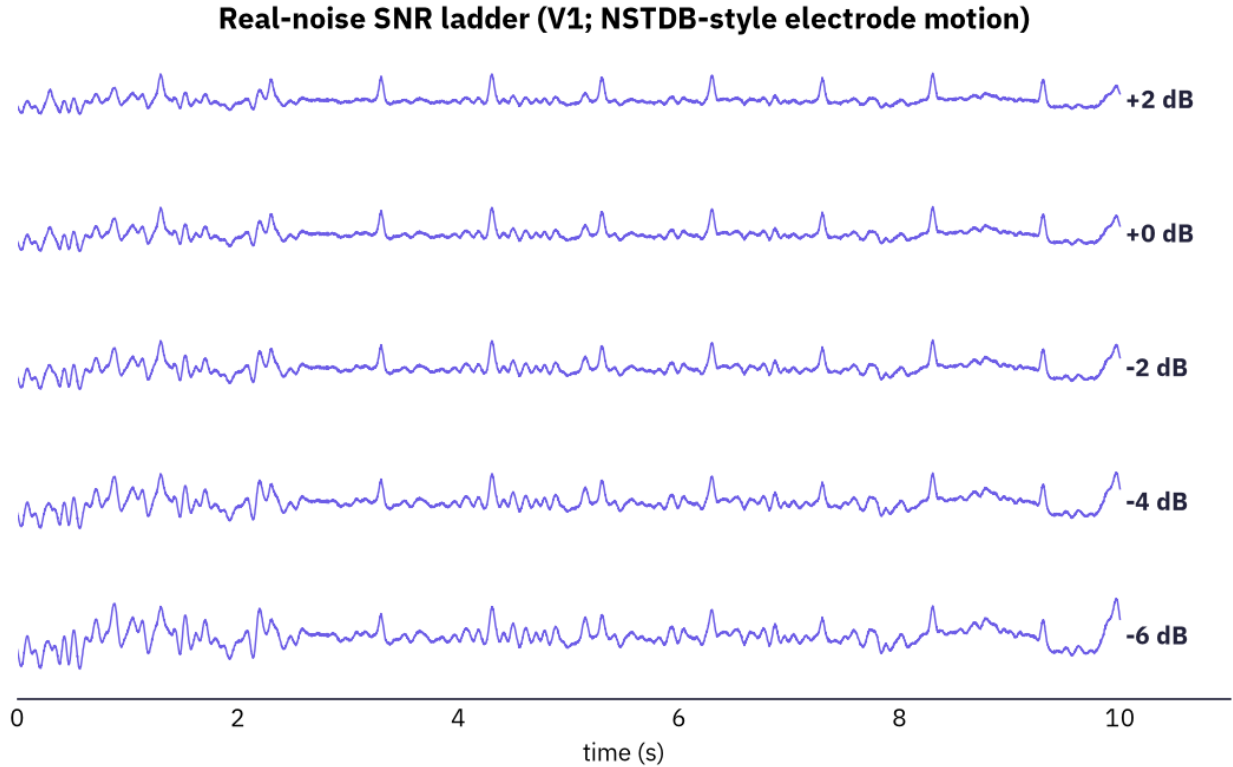


Figure 2: Real-noise SNR ladder (illustrative synthetic parent; NSTDB-style electrode motion).

4.3 Electrode-domain corruption. Limb-electrode potentials are recovered in the RA-as-reference gauge (RA = 0, LA = I, LL = II); chest-electrode potentials as $Ck = Vk + WCT$, $WCT = (RA+LA+LL)/3$. An artefact is added to a chosen electrode; the WCT and all twelve leads are recomputed, and the invalidated derived leads are recorded. (The classical WCT is an idealised electrode model, **not** a torso forward model.)

4.4 Engineering extremes. Deterministic builders for clipping/rail-saturation, flatline (zero), constant-ADC, lead-off, digital-missing (NaN), electrode-motion swings, step-and-exponential-

recovery, polarity inversion, and intermittent lead-off. NaN / flat / constant / rail cases are labelled `data_integrity_failure` with a type — **not** as noise.



Figure 3: Engineering extreme: an overload swing clipped at the acquisition rail.

5. Full record roster

All 85 records are listed below. **Expected quality** is the record-level `RecordQuality` a correctly-tuned gate should assign; **Discard** is the expected `noiseguard` record-level decision; **Bad leads** is the count of leads the label expects a detector to reject. PTB-XL parents are shown as record stems (the `records500/NNNNN/` directory prefix is omitted).

5.1 Naturally poor (15)

Real PTB-XL records selected by technical-validation quality flags; no clean parent (inherently bad).

Record	PTB-XL source	Quality flag(s)	Expected quality	Discard
<code>nlf_nat_001</code>	<code>10646_hr</code>	<code>electrodes_problems</code>	reject	yes
<code>nlf_nat_002</code>	<code>10652_hr</code>	<code>burst_noise</code>	limited	—
<code>nlf_nat_003</code>	<code>11176_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_004</code>	<code>12587_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_005</code>	<code>13777_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_006</code>	<code>14197_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_007</code>	<code>16149_hr</code>	<code>baseline_drift</code>	reject	yes
<code>nlf_nat_008</code>	<code>16981_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_009</code>	<code>17281_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_010</code>	<code>17293_hr</code>	<code>baseline_drift</code>	limited	—
<code>nlf_nat_011</code>	<code>19487_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_012</code>	<code>20478_hr</code>	<code>static_noise</code>	limited	—
<code>nlf_nat_013</code>	<code>00359_hr</code>	<code>static_noise</code>	reject	yes
<code>nlf_nat_014</code>	<code>06320_hr</code>	<code>baseline_drift</code>	limited	—
<code>nlf_nat_015</code>	<code>06968_hr</code>	<code>static_noise</code>	limited	—

5.2 Real-noise SNR-ladder pairs (30)

Clean PTB-XL parents mixed with NSTDB noise across the ladder $\{-6, -4, -2, +0, +2\}$ dB. Each ships with its untouched clean parent (*_clean).

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
nlf_noise_001	sinus	10236_hr	em @ -6 dB	12	12	reject	yes
nlf_noise_002	afib	10330_hr	ma @ -4 dB	6	6	limited	yes
nlf_noise_003	pac pvc	11089_hr	bw @ -2 dB	6	6	limited	yes
nlf_noise_004	bbb con-duction	11662_hr	em @ +0 dB	12	0	limited	yes
nlf_noise_005	st t	12040_hr	ma @ +2 dB	6	0	limited	—
nlf_noise_006	flutter svt	12086_hr	bw @ -6 dB	6	6	limited	yes
nlf_noise_007	sinus	12407_hr	em @ -4 dB	12	12	limited	yes
nlf_noise_008	afib	01306_hr	ma @ -2 dB	6	6	limited	yes
nlf_noise_009	pac pvc	13114_hr	bw @ +0 dB	6	0	limited	yes
nlf_noise_010	bbb con-duction	13332_hr	em @ +2 dB	12	0	limited	—
nlf_noise_011	st t	13611_hr	ma @ -6 dB	6	6	limited	yes
nlf_noise_012	flutter svt	01364_hr	bw @ -4 dB	6	6	limited	yes
nlf_noise_013	sinus	01382_hr	em @ -2 dB	12	12	limited	yes
nlf_noise_014	afib	14071_hr	ma @ +0 dB	6	0	limited	yes
nlf_noise_015	pac pvc	14211_hr	bw @ +2 dB	6	0	limited	—

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
nlf_ noise_ 016	bbb con- duction	14337_hr	em @ -6 dB	12	12	reject	yes
nlf_ noise_ 017	st t	15473_hr	ma @ -4 dB	6	6	limited	yes
nlf_ noise_ 018	flutter svt	15535_hr	bw @ -2 dB	6	6	limited	yes
nlf_ noise_ 019	sinus	15873_hr	em @ +0 dB	12	0	limited	yes
nlf_ noise_ 020	afib	15877_hr	ma @ +2 dB	6	0	limited	—
nlf_ noise_ 021	pac pvc	16615_hr	bw @ -6 dB	6	6	limited	yes
nlf_ noise_ 022	bbb con- duction	16930_hr	em @ -4 dB	12	12	limited	yes
nlf_ noise_ 023	st t	01709_hr	ma @ -2 dB	6	6	limited	yes
nlf_ noise_ 024	flutter svt	17553_hr	bw @ +0 dB	6	0	limited	yes
nlf_ noise_ 025	sinus	17727_hr	em @ +2 dB	12	0	limited	—
nlf_ noise_ 026	afib	17770_hr	ma @ -6 dB	6	6	limited	yes
nlf_ noise_ 027	pac pvc	17867_hr	bw @ -4 dB	6	6	limited	yes
nlf_ noise_ 028	bbb con- duction	18284_hr	em @ -2 dB	12	12	limited	yes
nlf_ noise_ 029	st t	18508_hr	ma @ +0 dB	6	0	limited	yes
nlf_ noise_ 030	flutter svt	19002_hr	bw @ +2 dB	6	0	limited	—

5.3 Engineering extremes — single-lead (13)

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_001_emg_v1	19238_hr	swing 6 mV p2p (V1)	—	1	limited	—
nlf_eng_002_macecg_walk_v2	19268_hr	MACECGDB walk swing 8 mV p2p (V2)	—	1	limited	—
nlf_eng_003_intermittent_v6	19661_hr	intermittent lead-off with reconnection (V6)	intermittent lead off	1	limited	—
nlf_eng_004_step_ii	01979_hr	baseline step 4 mV, tau=1 s (II)	—	0	limited	—
nlf_eng_005_macecg_jump_overload_v3	01995_hr	MACECGDB rail jump swing 30 mV p2p (V3) → rail ±5 mV	rail saturation	1	limited	—
nlf_eng_006_flatline_v4	20943_hr	flatline / zero-fill (V4)	flatline zero	1	limited	—
nlf_eng_007_constant_v5	20977_hr	stuck constant 1.2 mV (V5)	constant adc	1	limited	—
nlf_eng_008_missing_v6	21495_hr	digital-missing / NaN (V6)	digital missing channel	1	limited	—
nlf_eng_009_flatline_i	21820_hr	flatline / zero-fill (I)	flatline zero	5	limited	—
nlf_eng_010_leadoff_v2	21834_hr	lead-off (V2)	lead off	1	limited	—
nlf_eng_011_emg_burst_v5	02779_hr	swing 5 mV p2p (V5), t in [3,6] s	—	0	limited	—
nlf_eng_021_reversal_i	03941_hr	polarity inversion (I)	—	0	limited	—

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_031_flatline_ii	06175_hr	flatline / zero-fill (II)	flatline zero	5	limited	yes

5.4 Engineering extremes — multi-lead / mixed (27)

Electrode-domain propagation, precordial coupling, multi-lead saturation, and compound “wild” cases.

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_012_couple_v1v2	02965_hr	precordial coupling V1, V2 @ +0 dB	—	2	limited	—
nlf_eng_013_couple_v4v5	03016_hr	precordial coupling V4, V5 @ +6 dB	—	0	limited	—
nlf_eng_014_triple_v1v2v3	03047_hr	precordial coupling V1, V2, V3 @ -6 dB	—	3	limited	yes
nlf_eng_015_electrode_la	03365_hr	LA electrode motion 1 mV	—	4	limited	yes
nlf_eng_016_electrode_ra	03427_hr	RA electrode offset 3 mV	—	6	reject	yes
nlf_eng_017_saturate_v2v3	03543_hr	overload swing 30 mV p2p (V2, V3) → rail ±5 mV	rail saturation	2	limited	—
nlf_eng_018_saturate_v4v5v6	03687_hr	overload swing 30 mV p2p (V4, V5, V6) → rail ±5 mV	rail saturation	3	limited	yes
nlf_eng_019_electrode_ll	03730_hr	LL electrode motion 1.2 mV	—	3	limited	yes

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_ 020_ compound_ wild	03885_hr	LA electrode motion 0.8 mV ; MACECGDB walk swing 8 mV p2p (V2) ; lead-off (V5)	lead off	5	limited	yes
nlf_eng_ 022_ mixed_ integrity	03987_hr	digital- missing / NaN (V1) ; overload swing 25 mV p2p (V6) → rail ±5 mV ; baseline step 3 mV, tau=1.5 s (II)	digital missing channel	2	limited	—
nlf_eng_ 023_ saturate_ v1v2v3	04593_hr	overload swing 30 mV p2p (V1, V2, V3) → rail ±5 mV	rail saturation	3	limited	yes
nlf_eng_ 024_ saturate_ precordial_ all	00476_hr	overload swing 32 mV p2p (V1, V2, V3, V4, V5, V6) → rail ±5 mV	rail saturation	6	reject	yes
nlf_eng_ 025_emg_ v2v3_ severe	05141_hr	swing 12 mV p2p (V2, V3)	—	2	limited	yes
nlf_eng_ 026_ flatline_ v5v6	05267_hr	flatline / zero-fill (V5, V6)	flatline zero	2	limited	yes
nlf_eng_ 027_ constant_ v2v3	05298_hr	stuck constant 2 mV (V2, V3)	constant adc	2	limited	yes
nlf_eng_ 028_ missing_ v1v2v3	05883_hr	digital- missing / NaN (V1, V2, V3)	digital missing channel	3	limited	yes

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_ 029_ leadoff_ v4v5v6	05887_hr	lead-off (V4, V5, V6)	lead off	3	limited	yes
nlf_eng_ 030_ intermittent_ v2v5	05940_hr	intermittent lead-off with recon- nection (V2, V5)	intermittent lead off	2	limited	yes
nlf_eng_ 032_ electrode_ ra_severe	06526_hr	RA electrode offset 5 mV	—	6	reject	yes
nlf_eng_ 033_ electrode_ la_offset	06789_hr	LA electrode offset 4 mV	—	4	limited	yes
nlf_eng_ 034_ electrode_ ll_offset	07086_hr	LL electrode offset 4 mV	—	3	limited	yes
nlf_eng_ 035_ double_ electrode_ ra_ll	08699_hr	RA electrode offset 3 mV ; LL electrode offset 3 mV	—	6	reject	yes
nlf_eng_ 036_ baseline_ swing_ limb	08931_hr	MACECGDB walk swing 10 mV p2p (I, II)	—	2	limited	yes
nlf_eng_ 037_ compound_ wild2	09188_hr	LA electrode motion 1 mV ; overload swing 30 mV p2p (V3) → rail ±5 mV ; in- termittent lead-off with recon- nection (V6) ; baseline step 4 mV, tau=1.5 s (II)	rail saturation	5	limited	yes

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
nlf_eng_ 038_ mixed_ integrity2	09509_hr	digital- missing / NaN (V3) ; flatline / zero-fill (V5) ; stuck constant 1.5 mV (V1)	digital missing channel	3	limited	yes
nlf_eng_ 039_ reversal_ ii_v1	09836_hr	polarity inversion (II) ; polarity inversion (V1)	—	0	limited	—
nlf_eng_ 040_ catastrophic_ integrity	09915_hr	flatline / zero-fill (I) ; digital- missing / NaN (V6) ; overload swing 30 mV p2p (V3) → rail ±5 mV ; in- termittent lead-off with recon- nection (V4)	digital missing channel	8	reject	yes

6. Reproducibility and provenance

A single master seed (20260713) plus `provenance.json` (pinned source versions, per-record seed, source-file SHA-256) make the corpus bit-exactly regenerable:

```
make download      # NSTDB only (PTB-XL / PTB-XL+ / MACECGDB from the local mirror)
make select        # resolve PTB-XL parent ids -> recipes/source_ids/ (seeded)
make regenerate    # WFDB records + labels + manifest + provenance -> ./out
```

Determinism is per-record: `SeedSequence(entropy=master_seed, spawn_key=(record_index,))`. The corpus definition leaves source ids unresolved; `resolve_source_ids()` binds the seeded PTB-XL selection into the specs at generation time.

7. Verification

Confirmed end-to-end on the local PhysioNet mirror (PTB-XL 1.0.3, MACECGDB 1.0.0, NSTDB 1.0.0):

- **Counts:** 85 records (15 / 30 / 40) + 70 paired clean parents.
- **SNR ladder:** max |requested - measured| SNR error **0.0 dB** across all 30 mixes (tol 0.5 dB).
- **Data integrity:** digital-missing (NaN) round-trips through WFDB — the affected channel reads back all-NaN while intact leads stay finite.
- **Determinism:** two independent full regenerations are **byte-identical**.
- **Quality gates:** 74 unit/integration tests pass; **ruff** + **black** clean.

*Generated by Artefaux v2.0.0 (<https://github.com/depolarised/artefaux>) via `scripts/reports/generate_corpus_`.
every figure and table value is computed from the committed corpus definition.*